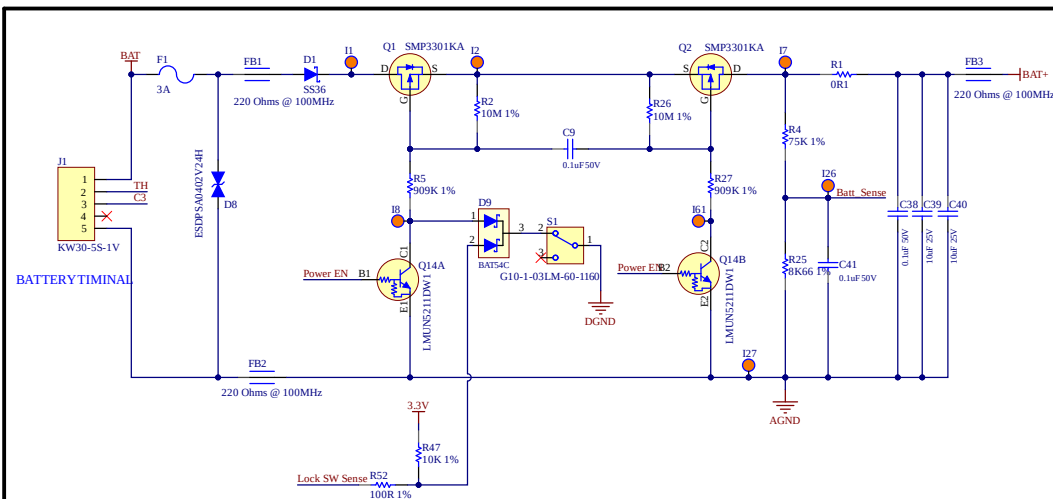
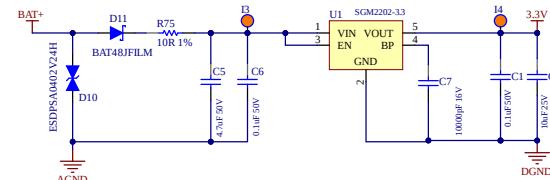
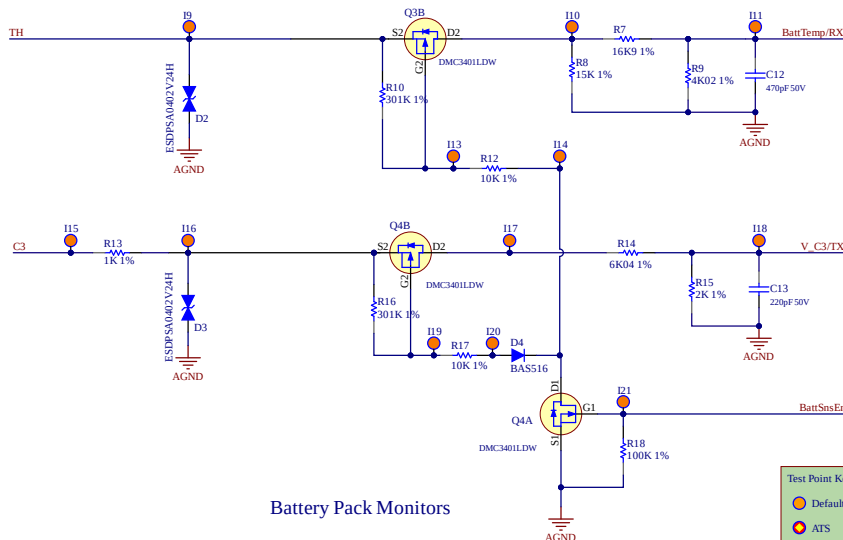




Lock SW and Reverse Protect Circuit



3.3V LDO



Battery Pack Monitors

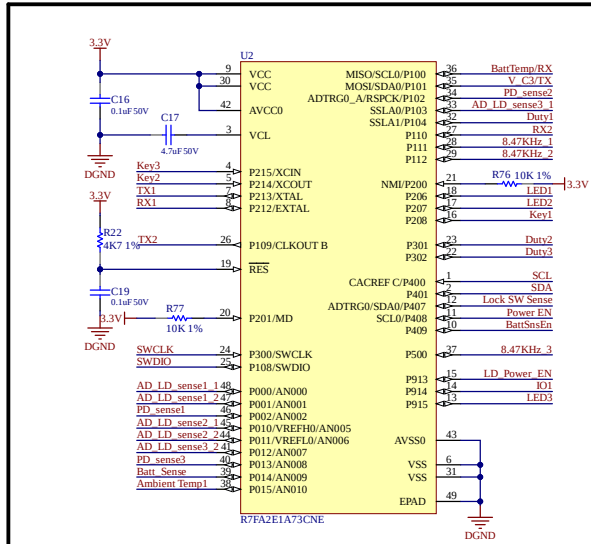
Test Point Key	
●	Default
●	ATS
●	Required
●	Optional
✕	Removed

GENERATED FROM AND CONTROLLED BY ALTUM	
PCB PART NUMBER:	NA837330
PCB REVISION:	B
SOURCE ITEM:	

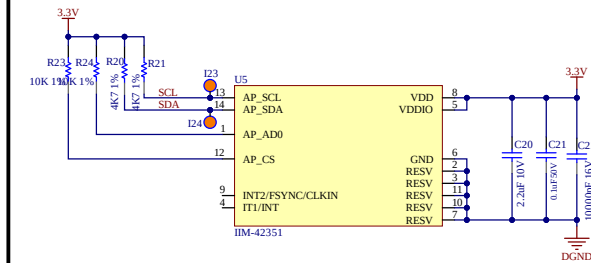
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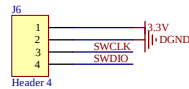
Time: 10:51:46 AM	DOC NO: N#####-SCH1
Date: 4/2/2026	TITLE: *** TITLE ***
Size: 11x17	*** TITLE 2 ***
Sheet: 1 of 3	



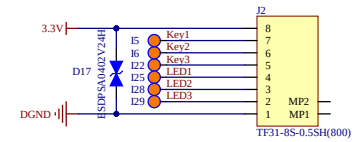
MCU and NTC



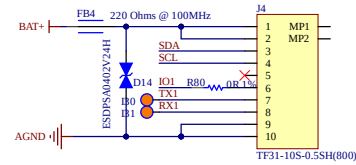
MEMS



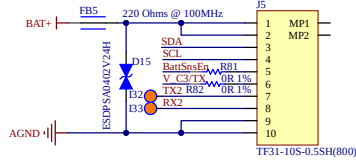
To Programmer



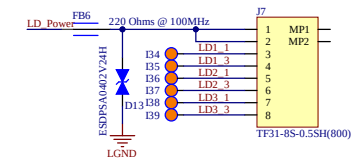
TACT SW AND LED CONNECTOR



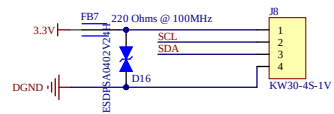
CONNECTOR 1(Reserved)



CONNECTOR 2(Reserved)



To Laser Driver Board

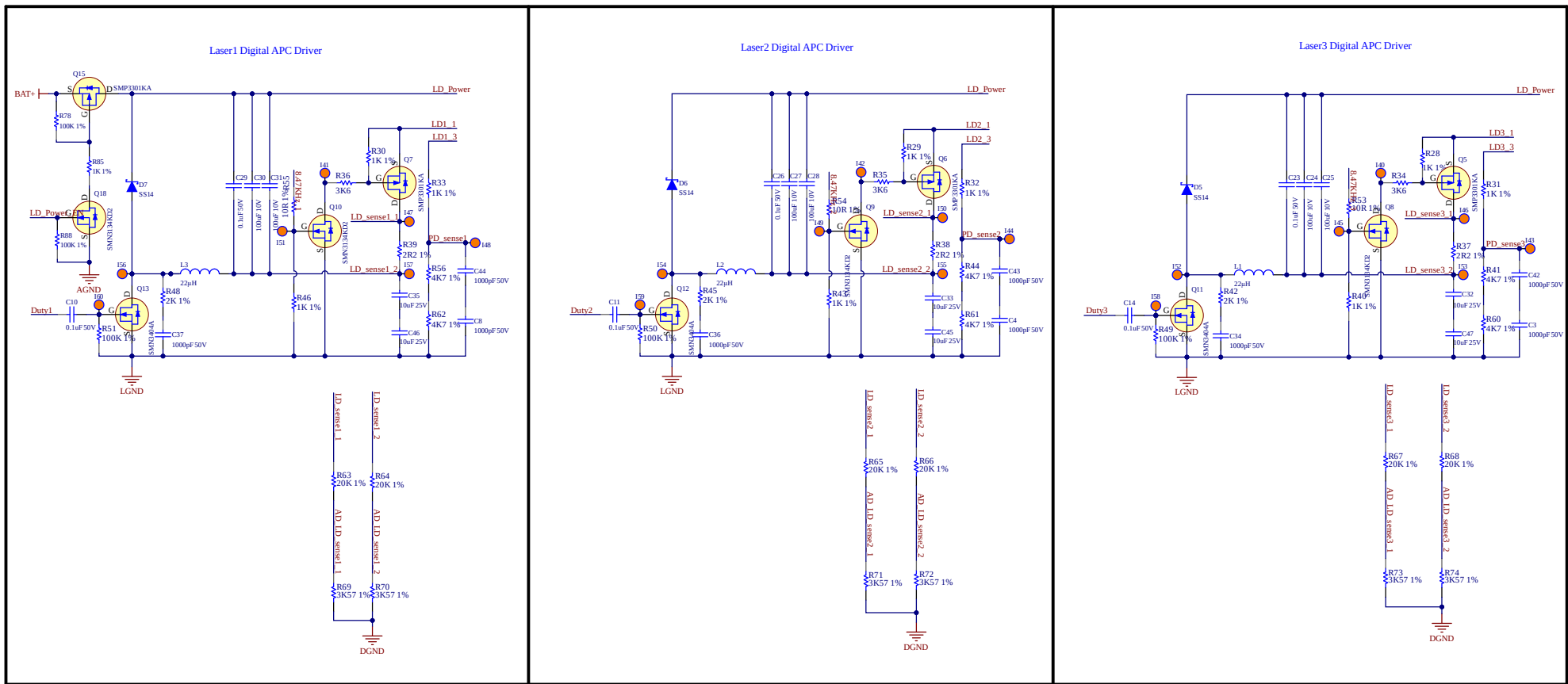


To MEMS Board

GENERATED FROM AND CONTROLLED BY ALTUM
PCB PART NUMBER: NAB37330
PCB REVISION: B
SOURCE ITEM:

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Date: 4/2/2026
Size: 11x17
Sheet: 2 of 3
DOC NO: N####-SCH1
TITLE: *** TITLE ***
*** TITLE 2 ***



GENERATED FROM AND CONTROLLED BY ALTUM

PCB PART NUMBER: NA837330

PCB REVISION: B

SOURCE ITEM:

BLACK & DECKER INC. 701 East Joppa Road
Towson, MD 21286

Time: 10:51:46 AM

Date: 4/2/2026

Size: 11x17

Sheet 3 of 3

DOC NO: N####-SCH1

TITLE: *** TITLE ***

*** TITLE 2 ***

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NOTES:

1. MILLIMETERS (mm) ARE THE CONTROLLING DIMENSIONS FOR THIS DRAWING.

2. FABRICATE PCB IN ACCORDANCE WITH THE LATEST REVISION OF IPC-6012 AND IPC-6011 CLASS 2, USING CUSTOMER SUPPLIED DATA FILES.

3. MATERIALS:

A. LAMINATE AND PREPREG (B-STAGE) TO BE IN ACCORDANCE WITH IPC-4101. MATERIAL MUST MEET UL 94V-0 FLAMMABILITY RATING.

☒ FR4 (135° C Tg min)

☐ FR4 (155° C Tg min)

☐ CEM-1

☐ CEM-3

B. COPPER FOIL TO BE IN ACCORDANCE WITH IPC-4562. SHALL PASS PEEL OF 10oz IN ACCORDANCE WITH IPC-TM-650. BASE COPPER WEIGHT:

OUTER: ☐ 0.5oz

INNER: ☐ 0.5oz

☒ 1oz

☐ 1oz

☐ 2oz

☐ 2oz

FINISHED COPPER WEIGHT:

OUTER: ☐ 0.5oz

INNER: ☐ 0.5oz

☒ 1oz

☐ 1oz

☐ 2oz

☒ 2oz

PLATED THRU HOLE MIN AVERAGE COPPER WALL THICKNESS:

☒ 20um (CLASS 1 & 2)

☐ 25um (CLASS 3)

C. LAYERS:

☐ SINGLE SIDED

☐ 2 LAYER

☐ 6 LAYER

☒ 4 LAYER

☐ 8 LAYER

D. THICKNESS (+/- 10%):

☐ 0.8mm (0.031")

☐ 2.4mm (0.093")

☐ 1.2mm (0.047")

☒ 1.6mm (0.062")

4. FINISH:

A. ALL EXPOSED CONDUCTIVE PATTERN AREAS NOT COVERED WITH SOLDER MASK OR OTHER PLATING SHALL BE

☐ HASL (HOT AIR SOLDER LEVELED)

☒ HALF (LEAD FREE HOT AIR SOLDER LEVELED)

☐ OSP (PLATED WITH ORGANIC SURFACE PROTECTANT)

☐ ENIG (ELECTROLESS NICKEL IMMERSION GOLD)

B. APPLY LIQUID PHOTO IMAGEABLE SOLDER MASK OVER BARE COPPER (SMOBC), PER IPC-SM-840.

☐ TOP SIDE ONLY

COLOR: ☒ GREEN

☐ RED

☐ BOTTOM SIDE ONLY

☐ BLUE

☐ BLACK

☒ BOTH SIDES

☐ WHITE

☐ YELLOW

C. SILKSCREEN SHALL BE PERMANENT, ORGANIC, NON-CONDUCTIVE INK. THERE SHALL BE NO SILKSCREEN ON ANY SOLDERABLE COMPONENT PAD.

COLOR: ☒ WHITE

☐ BLACK

☐ BLUE

☐ YELLOW

☐ RED

A. ALL INSIDE RADIUS SHALL BE NO MORE THAN 0,5MM

B. ALL BOARD GEOMETRY TO BE HELD TO +/- 0,2MM UNLESS OTHERWISE SPECIFIED.

6. VIA TYPES:

A. VIAS SPECIFIED AS TYPE 7 IN DRILL DRAWING SHALL BE VIA IN PAD PLATED OVER (VIPPO):

FILLING TYPE = CONDUCTIVE

B. VIAS SPECIFIED AS TYPE 1b IN DRILL DRAWING SHALL BE TENTED. MASK DATA MUST NOT BE MODIFIED.

7. TOLERANCES:

A. ALL HOLES SHALL BE LOCATED WITHIN 0,2mm DIAMETER OF TRUE POSITION.

B. LAYER TO LAYER REGISTRATION SHALL BE WITHIN +/- 0,1mm

8. MUST CONFORM TO LATEST REVISIONS OF ALL SPECIFIED STANDARDS.

9. BOARD MUST HAVE APPROVED MANUFACTURER'S UL IDENTIFICATION AND THE BOARD TYPE CODE INCLUDING A LEAD FREE SYMBOL ON SILK SCREEN.

10. B&D DEVELOPMENT ENGINEERING MUST APPROVE SAMPLES FROM PRODUCTION TOOLING PRIOR TO SHIPMENT OF PRODUCTION QUANTITIES.

11. MASTER ARTWORK AND FABRICATION DRAWING HELD BY B&D DEVELOPMENT ENGINEERING, MASTERS TO BE SENT TO B&D FOR APPROVAL PRIOR TO RELEASE TO BOARD VENDOR.

12. NO MODIFICATION TO CAM DATA WITHOUT PRIOR STANLEY BLACK AND DECKER ENGINEERING APPROVAL UNLESS PREVIOUSLY NOTED.

13. USE ONLY MATERIALS TO BE IN COMPLIANCE WITH THE LATEST ROHS/WEEE DIRECTIVE. CERTIFICATE SHALL BE ATTACHED ON SAMPLE ANALYSIS REPORT/SAR OR FPI.

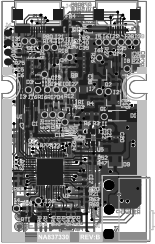
14. ALL MATERIALS MUST COMPLY WITH ES100118.

(NOTE REQUIRED BY STANLEY BLACK & DECKER; INTERNAL USE ONLY)

Multilayer Composite Print

4/2/2026

bl@stanley.com



NA837330

REV B

06FC6B01

NA837330

SUSP

REV B

YJH328N

101108

TOP ASSEMBLY

GENERATED FROM AND CONTROLLED BY ALTUM

PART NUMBER: NA837330

REVISION: B

SOURCE ITEM: 'ItemAndRevision'

Layer	Name	Material	Gerber
	Top Overlay		GT0
	Top Solder	Solder Resist	GTS
1	Top Layer		GTL
	Dielectric1	FR-4	
2	Mid-Layer 1		G1
	Dielectric2	FR-4	
3	Mid-Layer 2		G2
	Dielectric3	FR-4	
4	Bottom Layer		GBL
	Bottom Solder	Solder Resist	GBS
	Bottom Overlay		GB0

Total board thickness: 1.600mm

THIRD ANGLE PROJECTION



TOL UNLESS SPECIFIED

0 & 1 PLC DEC: ± 0.3

2 PLC DEC: ± 0.10

ANGLES: ± 1°

ALL DIMENSIONS mm

SIZE: 11x17

REF SCALE: NONE

SHEET: 105

Multilayer Composite Print

BLACK & DECKER INC.

DOC NO: NA837330-PCB1

TITLE: *** TITLE ***

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MASTER DRAWING

THIRD ANGLE PROJECTION

TOL UNLESS SPECIFIED

0 & 1 PLC DEC: ± 0.3

2 PLC DEC: ± 0.10

ANGLES: ± 1°

ALL DIMENSIONS mm

SIZE: 11x17

REF SCALE: NONE

SHEET: 105

Multilayer Composite Print

BLACK & DECKER INC.

DOC NO: NA837330-PCB1

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Comment	Description	Designator	Footprint	LibRef	Quantity
Cap0.1uF50V10% K6R042	CAPCER0.1uF50V 10%K6R042	C1, C8, C9, C10, C11, C14, C16, C19, C21, C23, C26, C28, C38, C41	CAPC1005060N_V2 [0402]	ECL-CAP-00006-V003	14
Cap10uF25V20% K6R003	CAPCER10uF25V 20%K6R003	C2, C32, C33, C35, C36, C38, C46, C48, C47	CAPC1608060N_V2 [0603]	ECL-CAP-00074-V002	9
Cap1000uF50V5% Q25D042	CAPCER1000uF50V 5%Q25D042	C3, C4, C8, C34, C36, C37, C42, C43, C44	CAPC1005060N_V2 [0402]	ECL-CAP-00017-V003	9
Cap4.7uF50V10% K7R063	CAPCER4.7uF50V 10%K7R063	C5, C17	CAPC1608060N_V2 [0603]	ECL-CAP-00770-V004	3
Cap10000uF16V10% K7R042	CAPCER10000uF16V 10%K7R042	C7, C22	CAPC1025060N_V2 [0402]	ECL-CAP-00012-V003	2
Cap470uF50V10% K7R042	CAPCER470uF50V 10%K7R042	C12	CAPC1005060N_V2 [0402]	ECL-CAP-00047-V003	1
Cap22uF50V10% K7R042	CAPCER22uF50V 10%K7R042	C13	CAPC1005060N_V2 [0402]	ECL-CAP-00037-V003	1
Cap2.2uF50V10% K7R063	CAPCER2.2uF50V 10%K7R063	C20	CAPC1608060N_V2 [0603]	ECL-CAP-00084-V002	1
Cap100uF10V20% K6R1206	CAPCER100uF10V 20%K6R1206	C24, C26, C27, C38, C39, C31	CAPC3216060N_V2 [1206]	ECL-CAP-00055-V001	6
IS06	DIODESCHOTTKY SS3660V3A 300L23W	D1	CPD1_N[600-123M]	ECL-DIO-00699-V001	1
ESDP9A02V24H	DIODE TVS3SDP9A042V24H 24V250W462	D2, D3, D6, D10, D13, D14, D15, D16, D17	Bi-Directional Diode_N [0402]	ECL-DIO-00692-V001	9
BA5B16	DIODES5B16V 250mA500E23	D4	SDOFL16065N[500- E23]_V2	ECL-DIO-00247-V003	1
SS14	DIODESCHOTTKY SS1440V1A 5MMVDC24AC	D5, D6, D7	DIOM6220G44L[DO- 24AC_5MM]	ECL-DIO-00356-V001	3
BA754C	DIODESCHOTTKY BA754C3N2000A SOT23 ARRLV	D8	SDT85P940120-3N [SOT-23-3]	ECL-DIO-00196-V002	1
BA78JFLM	DIODESCHOTTKY BA78JFLM40V 350mA500E23	D11	SDO2513X120N[SDO- E23]	ECL-DIO-00013-V005	1
F063943000V0327M	FUSEE063943000V0327M 1.0A125VDC3003F0327 BLCW	F1	FUSC1608065N[0603]	ECL-FUS-00087-V003	1
BLM85C221N1D	INDUCTOR0220 CHM2.2A0603	REL, REL, REL, REL, REL, REL, REL	RESC1608065N_V2 [0603]	ECL-IND-00104-V002	7
AW30-55-1V	CONNHEADERAW30-55 1V5PDS1MM3MT VERT	J1	HRODSE_HW30-55-1V	ECL-CON-00593-V002	1
TF31-85-0.5SH	CONNPCCTF31-85-0.5SH 8PDS0.5MM 9MTFRANGLE	J2, J7	HRODSE_TF31-85- 0.5SH	ECL-CON-00586-V001	2
TF31-10S-0.5SH	CONNPCCTF31-10S-0.5SH 10PDS0.5MM 9MTFRANGLE	J1, J5	HRODSE_TF31-10S- 0.5SH	ECL-CON-00582-V002	2
AW30-45-1V	CONNHEADERAW30-45 1V4PDS1MM3MT VERT	J8	HRODSE_HW30-45-1V	ECL-CON-00578-V004	1
FTC2010S2020MCA	IND220H0.85A1.1 CHM2520	L1, L2, L3	CJAN_FT252010S220 [DMCA2025H]	ECL-IND-00105-V001	3
SMF3030A	MOSEFTFCH SMF3030A33-5V3A SOT23	Q1, Q2, Q6, Q7, Q15	SDT85P940120GSD- 3N[SOT-23-3]	ECL-TRA-00499-V002	6
DMC3403LDW	MOSEFTNVPCH DMC3403LDW30V 800mA50T863	Q3, Q4	SDT85P210X10SGD- 6N[SC-70-6_SOT-363]	ECL-TRA-00021-V003	2
SMG1314K02	MOSEFTNCH SMG1314K02-20V 0.75AEN1006-2L	Q8, Q9, Q10, Q18	DN656P100055-3N [DFN1006-3]	ECL-TRA-00500-V001	4
SMG3404A	MOSEFTNCH SMG3404A30V4A SOT23	Q11, Q12, Q13	SDT85P940120GSD- 3N[SOT-23-3]	ECL-TRA-00458-V001	3
LMUN5211DW1	TRANSNPN LMUN5211DW150V 100mA50T863PRE- BAGED	Q14	SDT85P210X10ENBC- 6N[SC-70-6_SOT-363]	ECL-TRA-00190-V002	1
RES0R11%1206	RES01CHM1%1/2W 1206	R6	RESC3216060N_V2 [1206]	ECL-RES-01032-V004	1
RES10K1%1%0402	RES10KCHM1%1/16W 0402	R2, R26	RESC1005040N_V2 [0402]	ECL-RES-01575-V003	2
RES0R1%1%0603	RES0CHM1%1/12W 0603JAMR2	RL, R6	RESC1608065N_V2 [0603]	ECL-RES-00289-V006	3
RES75K1%1%0402	RES75KCHM1%1/16W 0402	R4	RESC1005040N_V2 [0402]	ECL-RES-00076-V006	1
RES900K1%1%0402	RES900KCHM1%1/16W 0402	R6, R27	RESC1005040N_V2 [0402]	ECL-RES-01916-V003	2
RES119K1%1%0402	RES119KCHM1%1/16W 0402	R7	RESC1005040N_V2 [0402]	ECL-RES-00102-V006	1
RES15K1%1%0402	RES15KCHM1%1/16W 0402	R8	RESC1005040N_V2 [0402]	ECL-RES-00098-V006	1
RES4021%1%0402	RES40KCHM1%1/16W 0402	R9	RESC1005040N_V2 [0402]	ECL-RES-00037-V006	1
RES301K1%1%0402	RES301KCHM1%1/16W 0402	R30, R36	RESC1005040N_V2 [0402]	ECL-RES-00031-V006	3
RES10K1%1%0402	RES10KCHM1%1/16W 0402	REL, REL, REL, REL, R24, R47, R76, R77	RESC1005040N_V2 [0402]	ECL-RES-00089-V006	8
RES1K1%1%0402	RES1KCHM1%1/16W 0402	REL, R28, R29, R30, REL, R32, R33, R40, REL, R46, R65	RESC1005040N_V2 [0402]	ECL-RES-00139-V006	11
RES96041%1%0402	RES604KCHM1%1/16W 0402	R54	RESC1005040N_V2 [0402]	ECL-RES-00095-V006	1
RES2K1%1%0402	RES2KCHM1%1/16W 0402	RES, R42, R45, R48	RESC1005040N_V2 [0402]	ECL-RES-00025-V006	4
RES100K1%1%0402	RES100KCHM1%1/16W 0402	REL, REL, REL, REL, REL, R86	RESC1005040N_V2 [0402]	ECL-RES-00088-V006	6
RES4071%1%0402	RES47KCHM1%1/16W 0402	R20, R21, R22, R41, R44, R56, R60, REL, R62	RESC1005040N_V2 [0402]	ECL-RES-00041-V005	9
RES9661%1%0402	RES866KCHM1%1/16W 0402	R25	RESC1005040N_V2 [0402]	ECL-RES-00095-V004	1
RES3665%1%0402	RES366KCHM5%1/16W 0402	R54, R55, R56	RESC1005040N_V2 [0402]	ECL-RES-00139-V005	3
RES2R21%1210	RES22CHM1%1/2W 1210	R27, R38, R39	RESC2225070N_V2 [1210]	ECL-RES-02164-V003	3
RES100R1%1%0402	RES100KCHM1%1/16W 0402	R62	RESC1005040N_V2 [0402]	ECL-RES-00087-V006	1
RES10R1%1%0402	RES10KCHM1%1/16W 0402	REL, R54, R55	RESC1005040N_V2 [0402]	ECL-RES-01188-V004	3
RES20K1%1%0402	RES20KCHM1%1/16W 0402	REL, R64, R65, R66, REL, R69	RESC1005040N_V2 [0402]	ECL-RES-00832-V004	6
RES3671%1%0402	RES337KCHM1%1/16W 0402	R68, R70, R71, R72, R73, R74	RESC1005040N_V2 [0402]	ECL-RES-01797-V003	6
RES10R1%1%0603	RES10KCHM1%1/10W 0603	R75	RESC1005065N_V2 [0603]	ECL-RES-00885-V003	1
RES0R1%1%0402	RES0CHM1%1/12W 0402JAMR2	R80, REL, R82	RESC1005040N_V2 [0402]	ECL-RES-01312-V003	3
TSMDA103F36D1	THERMNTIC10KCHM 1%0603	R71	RESC1608065N_V2 [0603]	ECL-THER-00007-V007	1
G10-1-03M-60-1180	SWITCHG10-1-03M-60- 1180 PUSHBUTTON SPOT 1A125VTH	SL	G10-1-03M-60-1180- Footprint-1	ECL-SW-00014-V013	1
SCM202- 3.3VW65G/TR	ICREG50A202-3.3 LDO3.3V0.15ASOT23-3	U1	SDT85P9801045-SN [SOT-23-3]	ECL-IC-00041-V002	1
RTFA2ELAT73CNE	ICMCU RTFA2ELAT73CNE 128T144KBFLASH HWCF5-48	U2	QFN637000X0080-48- PQFN04048C_A,N	ECL-IC-00573-V004	1
IM442851	ICACCELLCOMPBR IM442851-3-AVS SC2023N LG44-14	U5	TDK[IM4285]_J[GA14]	ECL-IC-00085-V002	1